

ChatGPT

Jensen-Shannon Divergence

Jensen-Shannon (JS) Divergence.	2
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The **Jensen-Shannon Divergence (JSD)** is a method to measure **how different two probability distributions are**.

It is based on Kullback-Leibler Divergence but fixes some of its major limitations.

Intuition

Suppose you have two probability distributions:

- $P \rightarrow$ real distribution
- $Q \rightarrow$ predicted/model distribution

JS Divergence answers:

“How much information is lost when we treat these two distributions as similar?”

You can think of it as a **smoothed and symmetric version of KL Divergence**.

Why JS Divergence Was Needed

KL divergence has problems:

KL is asymmetric

$$D_{KL}(P||Q) \neq D_{KL}(Q||P)$$

So distance changes depending on order.

KL can become infinite

If:

$$Q(x) = 0 \quad \text{but} \quad P(x) > 0$$

then:

$$\log \frac{P(x)}{Q(x)} \rightarrow \infty$$

This is problematic in ML.

Core Idea of JS Divergence

Instead of directly comparing P and Q ,

JS divergence:

1. Creates an average distribution:

$$M = \frac{1}{2}(P + Q)$$

2. Measures how far both distributions are from this average.

Mathematical Formula

The Jensen-Shannon divergence between distributions P and Q :

$$JS(P \parallel Q) = \frac{1}{2}D_{KL}(P \parallel M) + \frac{1}{2}D_{KL}(Q \parallel M), \quad M = \frac{1}{2}(P + Q)$$

where:

- D_{KL} = KL divergence
- M = midpoint/average distribution

Expanding the Formula

Recall KL divergence:

$$D_{KL}(P \parallel Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

So JS divergence becomes:

$$JS(P \parallel Q) = \frac{1}{2} \sum_x P(x) \log \frac{P(x)}{M(x)} + \frac{1}{2} \sum_x Q(x) \log \frac{Q(x)}{M(x)}$$

Step-by-Step Example

Suppose:

$$P = [0.8, 0.2]$$

$$Q = [0.4, 0.6]$$

Step 1: Compute Average Distribution

$$M = \frac{1}{2}(P + Q)$$

$$M = [0.6, 0.4]$$

Step 2: Compute $KL(P || M)$

$$D_{KL}(P||M) = 0.8 \log \frac{0.8}{0.6} + 0.2 \log \frac{0.2}{0.4}$$

Approx:

$$= 0.8(0.288) + 0.2(-0.693)$$

$$= 0.2304 - 0.1386$$

$$\approx 0.0918$$

Step 3: Compute $KL(Q || M)$

$$D_{KL}(Q||M) = 0.4 \log \frac{0.4}{0.6} + 0.6 \log \frac{0.6}{0.4}$$

Approx:

$$= 0.4(-0.405) + 0.6(0.405)$$

$$= -0.162 + 0.243$$

$$= 0.081$$

Step 4: Final JS Divergence

$$JS(P||Q) = \frac{1}{2}(0.0918) + \frac{1}{2}(0.081)$$

$$\approx 0.0864$$

So the distributions are somewhat different but not extremely far apart.

Properties of JS Divergence

1. Symmetric

$$JS(P||Q) = JS(Q||P)$$

Huge advantage over KL.

2. Always Finite

Even if one distribution contains zeros.

Because M smooths the distributions.

3. Bounded

Using log base 2:

$$0 \leq JS(P||Q) \leq 1$$

- $0 \rightarrow$ identical distributions
 - $1 \rightarrow$ maximally different
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4. Metric Version Exists

Square root of JS divergence:

$$\sqrt{JS(P||Q)}$$

is a true mathematical distance metric.

Geometric Interpretation

Imagine probability distributions as points in space.

- KL divergence measures directed information loss.
- JS divergence measures how differently both distributions behave relative to their shared center.

So JS is more like a balanced similarity score.

Relationship with Entropy

JS divergence can also be written using entropy:

$$JS(P \parallel Q) = H\left(\frac{P+Q}{2}\right) - \frac{1}{2}H(P) - \frac{1}{2}H(Q)$$

where:

- $H(P)$ = entropy of P

This form explains the name:

- "Jensen" → from Jensen's inequality
- "Shannon" → from Shannon entropy

JS Divergence vs KL Divergence

Feature	KL Divergence	JS Divergence
Symmetric?	No	Yes
Infinite possible?	Yes	No
Stable in ML?	Less	More
Range bounded?	No	Yes
Metric form exists?	No	Yes

Applications in Machine Learning

1. GANs (Generative Adversarial Networks)

Original Generative Adversarial Network training minimizes JS divergence between:

- real data distribution
- generated distribution

2. NLP / Topic Modeling

Used to compare:

- word distributions
- topic distributions
- sentence embeddings

3. Clustering

Measures similarity between probability distributions.

4. Bioinformatics

Used in gene expression comparison.

Why GANs Had Problems with JS Divergence

When distributions do not overlap:

$$JS(P||Q)$$

becomes nearly constant.

Then gradients vanish.

This led to alternatives like:

- Wasserstein Distance
- Earth Mover's Distance

in Wasserstein GAN.

Simple Analogy

Imagine:

- P = music taste of person A
- Q = music taste of person B

JS divergence:

1. Creates an "average listener"
2. Measures how different each person is from that average
3. Combines both differences fairly

That fairness and symmetry are why JS divergence is widely used.